

# Dog Breed Identification Using Transfer Learning

## Project Report

### Abstract

Dog breed identification is an important task in computer vision. Transfer learning allows pre-trained CNN models like MobileNet, ResNet50, and VGG16 to classify dog breeds efficiently. This project uses transfer learning to improve accuracy and reduce training time.

### Introduction

Dog breed identification is a fine-grained image classification problem. Transfer learning helps reuse knowledge from models trained on large datasets like ImageNet.

### Objectives

- Build a dog breed classifier
- Use transfer learning
- Improve accuracy
- Reduce training time
- Predict dog breed from image

### Methodology

Steps include dataset collection, preprocessing, transfer learning, model training, and evaluation.

### Tools and Technologies

Python, TensorFlow, Keras, NumPy, OpenCV, Matplotlib, Scikit-learn

### Results

Transfer learning achieved high accuracy and reduced training time.

### Applications

Veterinary clinics, animal shelters, pet apps, and security systems.

## **Conclusion**

Transfer learning provides an efficient and accurate solution for dog breed identification.